

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 4

8 Cybersecurity (fms27)

- (a) Compare and contrast, under at least four relevant aspects, the mechanical implementation of a master-keyed lock system against one using electronic key fobs. Then give one scenario where you would recommend the mechanical implementation and one where you would recommend the electronic one, justifying why. [6 marks]

---

*Answer:*

*Here and elsewhere: these are solution notes for the benefit of checkers and examiners, not model answers, and should not be taken as indicative of the amount of text a candidate ought to write in order to secure full marks.*

- **Reconfigurability** The electronic version can be reprogrammed at will to support different sets of keys without having to re-pin each lock. This is particularly advantageous with a dynamic population of users, where issued keys must periodically be revoked.
- **Privacy/traceability** With the electronic version it is possible to log each occurrence of a lock being opened (and possibly even each *attempt* at opening a lock, successful or not, logging the key used). This is an advantage for traceability and, conversely, a disadvantage for privacy.
- **Reliability** Mechanical locks have fewer parts that can go wrong, do not have software that may contain bugs and do not require electricity to run.
- **Number of key differs** With electronic locks the number of possible key differs is essentially unlimited and in any case much greater than with mechanical locks.
- **Cost** For an electronic lock deployment with many locks and with frequent reprogramming or reissuing of key fobs (as opposed to a stand-alone electronic lock for a single door), a back-end software system will also be needed, as well as some form of networking (sneakernet at minimum; wired or wireless in fancier cases) between the locks and the back-end. All this will make the electronic lock system more expensive to purchase and deploy. On top of that, it is likely that the software system will need some form of subscription or maintenance contract, meaning additional recurring costs not usually present in the case of a mechanical system.
- **Versatility** The mechanical lock is physically limited in the schemes it can implement; indeed it even introduces undesirable additional keys, as explored in part (b)(ii). The electronic lock, instead, can do anything one programs it to do.

*(One mark per sensible dimension, up to 4.)*

I would recommend a mechanical system in a small-scale residential setting, with no requirements for frequent re-keying, where reliability and privacy were significant concerns and where the initial and recurring costs of the electronic solution would not be acceptable.

I would recommend an electronic system for a large hotel, with guests changing on a daily basis, where frequent revocation of keys is a necessity and traceability a desirable feature.

*(One mark for each sensibly justified scenario, up to 2.)*

---

- (b) A building with 24 door locks, D0–D23, uses a mechanical (pin-tumbler) master key system. Locks have 5 pin stacks, there are 8 possible cut depths per stack from 0 (shallowest) to 7 (deepest) and each stack contains at most one master pin. Pretend, unrealistically, that all cut depths are usable. Each door has its

own differ key that opens just that door. Doors D10–D19 may also be opened by master key 36735. All doors, D0–D23, may also be opened by grand master key 12735.

- (i) In the lock for door D12, what is the minimum and maximum possible number of pin stacks containing a master pin? Explain your reasoning. [2 marks]

---

*Answer:* We know this door is opened by three different keys: the differ key for D12, the master key for D10–D19 and the grand master key for D0–D23. Each pin stack containing a master pin multiplies by two the number of keys that open that lock. Thus, with 0 master pins, only one key opens the lock; with 1 master pin, only 2 keys; with  $m$  master pins,  $2^m$  keys. Since the smallest  $m$  for which  $2^m \geq 3$  is  $m = 2$ , that is the required minimum. There are no constraints on the maximum: it would all still work if all pin stacks had a master pin, hence the maximum is 5, the number of pin stacks. (1 mark for correct min and max, 1 mark for meaningful explanation.)

---

- (ii) Door D13 is opened by differ key 32737. List *all* the other keys that also open that door. Mention and explain your assumptions. [4 marks]

---

*Answer:* Let's align the working keys that we already know about:

DK    32737  
MK    36735  
GMK   12735

By comparing the above keys column by column, we know for sure that at least pin stacks 1, 2 and 5 have a master pin (of depth 2, 4, 2 respectively), meaning this lock can be opened by  $2^3 = 8$  different keys, which are  $\{1,3\}$ - $\{2,6\}$ -7-3- $\{5,7\}$  or, exhaustively,

12735 (GMK)  
12737  
16735  
16737  
23735  
32737 (DK)  
36735 (MK)  
36737

(2 marks for writing out the 5 other keys. If not done, 1 mark for identifying the master pins and their correct depths.)

What we cannot say either way with the information given is whether pin stacks 3 and 4 have a master pin or not. The three keys we witnessed give no evidence of that. (2 marks.) If the implementers of the system valued simplicity (and security!) there would be no other master pins, but we cannot say for sure—especially given that they are already using 3 master pins where 2 would have sufficed, as we established in part (b)(i). However, if there were more master pins, each of them would multiply by two the number of keys that open this lock (16 keys for 4 master pins, 32 keys for 5 master pins), though we'd need to figure out the position of the other cuts to be able to list the actual key codes.

---

- (iii) The occupant of the D16 office, who legitimately owns the D16 differ key 32535 but not the master nor the grand master key, wants to open door

D07 using the Matt Blaze privilege escalation attack, visiting door D07 only once in order not to arouse suspicion. She processes the pins left to right, processes the pin depths from shallowest to deepest and uses an optimal strategy. Assume there are no master pins in the D16 lock other than those necessary for it to be opened by the indicated differ key, master key and grand master key. List, in order, every key that the attacker must create and try, mentioning which lock she tries it on and whether it opens, and giving a clear running explanation of what she does. How many keys exactly (not an upper bound) does she need to test on D16, and how many on D07, before successfully opening D07? [8 marks]

---

*Answer:* She starts with 5 blanks that we'll call  $P_1$  to  $P_5$ . She files each blank  $P_i$  to match her D16 differ key in all positions except  $i$ , where she initially leaves it at 0.

Then, for each  $P_i$  key, she tries all possible depths  $d \in \{0..7\}$  in pin position  $i$ , checking every time whether the key opens the D16 lock. We'll call these variant keys  $P_{i,d}$ . Physically it's still the same key  $P_i$  but it gets filed to progressively deeper positions. Small optimisations: no need to probe the depth that already exists in the D16 differ key, and no need to probe further depths after finding two cuts in the same pin stack.

For pin stack 1,  $P_{1,0} = 02535$ . She tries it in D16 (does not open); she progressively files down the leftmost position (pin stack 1) to  $P_{1,1} = 12535$  and immediately finds it works. She deduces the cuts are at 1 (found) and 3 (already on her differ key), and stops there. No need to cut this key any further. **Keys tried: 2.**

For pin stack 2,  $P_{2,0} = 30535$ . She tries the depths from 0 to 7 in that stack, omitting 2 because it's in her own differ key, but stops at 6 which opens. Thus the cuts in stack 2 are at depths 2 and 6 and she **tried 6 keys**.

For pin stack 3,  $P_{3,0} = 32035$ . She tries depths 0 to 7 (which works) except 5 (in her differ key). Depth 7 works. Cuts: at 5 and 7. **Keys tried: 7.**

For pin stack 4,  $P_{4,0} = 32505$ . She tries all depths from 0 to 7 except her own (3) and finds that none of them opens D16. There is no master pin in position 4. Cuts: at 3. **Keys tried: 7.**

For pin stack 5,  $P_{5,0} = 32530$ . She tries all depths from 0 to 7 except her own (5) and finds there is no master pin. Cuts: at 5. **Keys tried: 7.**

From the above  $2 + 6 + 7 + 7 + 7 = 29$  **key trials**, all on D16 which safe for her, she knows that the D16 lock has valid cuts at  $\{1,3\}$ - $\{2,6\}$ - $\{5,7\}$ -3-5.

From this she now generates the 8 possible keys (or rather 7 excluding her own) that open D16, namely

12535  
12735  
16535  
16735  
32535 (not generated; own differ key)  
32735  
36535  
36735

She knows for sure that one of them must be the grand master key. She then goes to D07 to try these 7 keys in sequence (her own differ key opens D16 but cannot be the grand master key), but obviously without bothering to try any further keys as soon as one of them opens the door. We happen to know (from the problem statement of part

— *Solution notes* —

(b)) that the grand master key is 12735 so she'll be successful at her **second** attempt. **Keys tried on D07: 2.** Of course the 2 is pure luck because of ordering; worst case, it would have been 7 tries.

*Not required for full marks, but for reference here is the sequence of keys she produces and tests on D16.*

Pin stack 1: \*

P10 02535

P11 12535 WORKS; cuts @ 1,3; no need to continue with P1x.

Pin stack 2: \*

P20 30535

P21 31535

P22 32535 (not tried: own differ key)

P23 33535

P24 34535

P25 35535

P26 36535 WORKS; cuts @ 2,6; no need to continue with P2x.

Pin stack 3: \*

P30 32035

P31 32135

P32 32235

P33 32335

P34 32435

P35 32535 (not tried: own differ key)

P36 32635

P37 32735 WORKS; cuts @ 5,7

Pin stack 4: \*

P40 32505

P41 32515

P42 32525

P43 32535 (not tried; own differ key)

P44 32545

P45 32555

P46 32565

P47 32575

no master pin here; only cut is 3

Pin stack 5: \*

P50 32530

P51 32531

P52 32532

P53 32533

P54 32534

P55 32535 (not tried; own differ key)

P56 32536

P57 32537

no master pin here; only cut is 5

---